

Brain-Drive: A Smart Driver for Controlling Digital Appliances Using Cognitive Command

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Abstract—for the disabled and handicapped, the world of brain-computer interface has opened a new horizon to lead an unassisted and least supervised life. But the use of prosthetic limbs and artificial organs are still not commercially available in most of the countries. Recently developed some other brainwave based systems are mostly expensive, immobile or task specific. In order to overcome these problems, analogous to the pen-drive, we have focused on developing a low cost control unit namely “Brain-drive” which transforms ones thoughts into output digital electric signal. Certain thoughts (mental tasks) with hard eye blinking (neural driven physical outcome) develops certain action potential, no matter the eye blinking is visible or not, it creates a significant change in electroencephalogram (EEG) or brain signal. Single channel EEG signal is collected from frontal lobe as it plays a vital role in voluntary movement using a wearable Mindwave Mobile from Neurosky. The EEG data thus recorded is preprocessed to reduce the effect of noise and artifacts and then analyzed in time domain. A grid of cells are displayed on a monitor in front of the subject, which is connected to individual digital output pin onboard an embedded system. To select a cell, a user just needs to look and generate a sustainable eye blink which can easily be identified from the raw value of brainwave. Since eye blinking can be both conscious and involuntary, automatic periodic eye blink of very low frequency oscillation is removed to avoid unwanted selection of command. Unlike webcam based eye blink detector, Brain-Drive captures the signal for blink rather than the visual output of blinking. The system can be trained for person specific customized performance, considering special needs of a disabled or handicapped people who cannot communicate with the outside world using voice, motion or other methods. With the help of cognitive command, the ‘Brain-Drive’ can be used to control wheelchair, home appliances and even industrial machines.

Keywords—*Electroencephalogram(EEG), Brainwave, Brain Computer Interference(BCI)*

I. INTRODUCTION

Right from the very beginning of the arrival of computer system, it was essential that some kind of method existed that

allows human to commutate with the machines. These communication systems proved so important that sometimes the success of the entire device pivoted upon how user friendly, effective and efficient the communication system was. These communication mechanisms travelled a long way the past decade from the primitive keyboard controls to the more advanced touch screen, voice activated and gesture input systems. Thanks to the current advances in cognitive neuroscience and neuroimaging technologies, it is now possible to initiate direct communication between the human brain and the machines. The ability to control equipment with the brain has opened up a whole new world of possibilities. Although there are numerous possible applications like virtual reality kits, room control systems, wireless headphones etc. but the most effective uses lies with handicapped people. Specially, those who suffer from partial or full body paralysis with their brain being the only fully functioning organ available. With efficient use of such control systems, even full body paralyzed patients can independently control a wide array of different systems ranging from room control equipment like fan and light to fully equipped computer systems. The brain control system can also be hooked up with smart phones to contact authorities in case of emergencies.

Electroencephalography (EEG) is the measurement of electrical activity in the living brain. Nowadays in clinical use of EEG, 21 electrodes are used, which costs thousands of dollars. In recent years, inexpensive mobile EEG devices have been developed by Avatar EEG Solutions, Neurosky, OCZ Technology, InteraXon, PLX Devices and Emotiv Systems. These devices are not used in clinical use, but are used in the Brain Control Interface (BCI) and neuro feedback (one of biofeedback type). The BCI is a direct communication pathway between the brain and an external device. One of the cheapest EEG device is single-channel MindWave headset produced by Neurosky Inc., San Jose, CA. [1]

TABLE 1. WAVES GENERATED FOR DIFFERENT TYPES OF ACTIVITY IN THE BRAIN

Brainwave Type	Frequency (Hz)	Mental States and Condition
Delta	0.1-3	Deep, dreamless sleep, non-REM sleep, unconscious
Theta	4-7	Intuitive, creative, recall, fantasy, imaginary, dream
Alpha	8-12	Relaxed (but not drowsy) tranquil, conscious
Low Beta	12-15	Formerly SMR, relaxed yet focused, integrated
Midrange Beta	16-20	Thinking, aware of self & surroundings
High Beta	21-30	Alertness, agitation

Recently, a lot of works have been devoted towards brain communication systems. Especially, extraction and manipulation of the EEG signals are getting popularity in various applications, such as controlling a mobile robot using noninvasive extraction of EEG signal [2], advanced video game control platform using EEG [3], and automation systems using brain machine interface [4]. Other works like wheelchair control using EEG [5] and machine control using EEG [6] have received much attention in recent times. Moreover, extraction, de-noising and artifact removal of EEG signal have also received considerable attention [7]. But most of them concentrate on a specific application or user specific functions, while building a standalone and application independent device is still in great demand for ease of its adoption in multifarious day to day life tasks.

The objective of this research is to develop a user friendly EEG based efficient brain-computer interfacing system which can easily be utilized to drive varieties of devices. The proposed standalone system extracts EEG data from the user and can generate eight different digital signals which can be hooked up to any kind of devices. The user-friendly interface then allows the user to control the hooked up devices with the brain alone. Choice of possible devices may include household appliances like lights and fans, emergency texts to ambulance or fire service, basic computer controls and many others.

II. PROPOSED ALGORITHM

A. Signal Acquisition

Electrical signals generated by neuron firing in the brain are the main controlling signal of the proposed model. The patterns and frequencies of these electrical signals can be measured by placing a sensor on the scalp. For this purpose MindWave Mobile from Neurosky has been used [8]. The

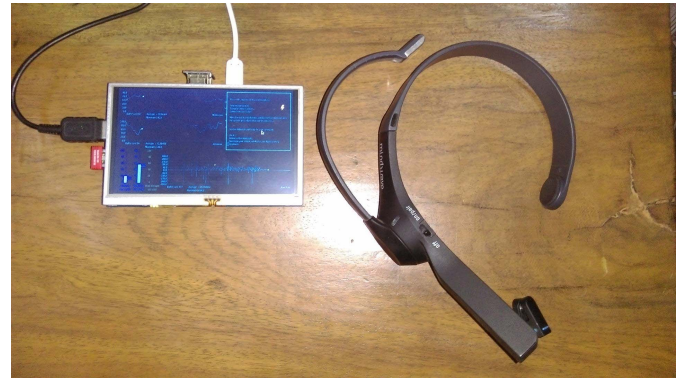


Fig. 1 Full Setup of the System

MindTools line of headset products contain NeuroSky Gear™ technology, which measures the analog electrical signals, commonly referred to as brainwaves, and processes them into digital signals. The ThinkGear technology then makes those measurements and signals available to applications using Bluetooth. TABLE 1 gives a general synopsis of some of the commonly-recognized frequencies that tend to be generated by different types of activity in the brain [9].

To provide portability to the whole system, Raspberry Pi 3 is being used as the processing unit. The Raspberry Pi is a small single-board computer. The Raspberry Pi 3 is the newest edition of the Raspberry Pi series and belongs a quad-core Cortex-A53 processor which is fast enough to process our signal. It also has a built in Bluetooth which we used to pair with Mindwave mobile.

Preliminary development board containing switching circuits has been developed for demonstration purpose. The development board receives digital signal directly from the GPIO of Raspberry Pi. There are total 26 GPIO pins in Raspberry Pi 3.

The full setup has been shown in Fig. 1. In this figure, the EEG sensor unit is shown on the right side (like a head set) and the screen side of the Raspberry Pi 3 unit is shown on the left side.

B. System Activation

MindWave Mobile sends a packet of data containing values of poor signal quality, attention, meditation and some raw wave values continuously to the processing unit via Bluetooth [10]. We have just used the attention, meditation and the raw brainwave data to control our system. The received data are pursued according to datasheet provided and values of attention, meditation and raw brainwave are recorded.

Raspberry Pi 3 processes the stored data once every second. Either attention or meditation value is used for activating our proposed system first. Different person has different degree of control over their consciousness. Some can control their mediation value while others can control attention value better than the other. That's why user has been given a freedom to select the system activation method either by attention or by meditation mode.



Fig. 2 Display of raw attention and meditation values stored with respective threshold.

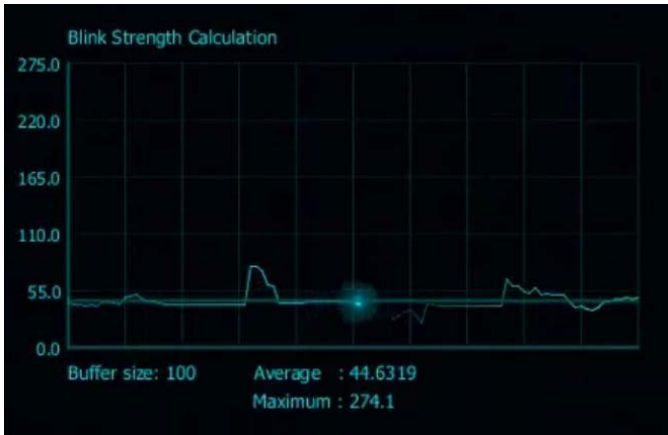


Fig. 3 Display corresponding to blink strength calculation.

Based on the choice of user, the process calculates the local average (LA) of last M samples ($M = 5, 10, 15, \dots$) of the attention or meditation values stored.

At every n samples ($n = 60, 70, \dots$) the system calculates the global average (GA), global maximum (max), global minimum (min) of attention or meditation and mean square of the raw value. It divides the interval between maximum and minimum into m steps ($m = 5, 10, \dots$) and sets a threshold by using the following formula

$$\text{Threshold } \delta_1 = (\max - \min) \times w_1 + \min \quad (1)$$

where w_1 is the weight preset by the programmer and ranges from 0 to 1. In Fig. , raw attention and meditation values with respective threshold in straight line have been shown.

The process then compares the local average (LA) with the calculated threshold and if it is above the calculated threshold, the input sequence is activated otherwise the system stays in the idle mode.

C. Blink Detection

For blink detection we propose to use the short-time-energy of the raw signal. The short time energy is given by

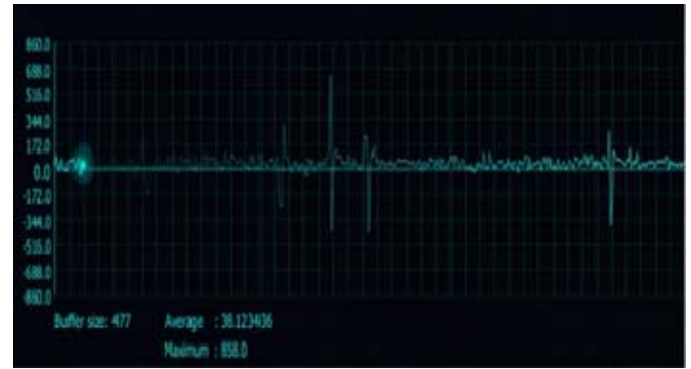


Fig. 2 Plot of Raw brainwave

$$P = \frac{1}{N} \sum_{m=1}^N |x(m)|^2 \quad (2)$$

where N is the window length and $x(m)$ is the value of the m^{th} sample of the brainwave signal. In Fig. 3, blink strength calculation is depicted.

Whenever the user makes a strong blink, a spike with high amplitude is detected in short time energy. We have used this feature to detect a blink. The amplitude of the short-time-energy of the raw signal gives the strength of the blink. An average of every M values is considered as the moving average value and it is being updated every time after P is calculated. Threshold to determine blink is updated according to (3).

$$\text{Threshold } \delta_2 = (\max - \min) \times w_2 + \min \quad (3)$$

where w_2 is the weight parameter defined by the program which ranges from 0 to 1.

The strength of the blink is given by the magnitude of short term energy P. A hard blink is detected if the short term energy follows the following condition

$$P \geq \delta_2 \quad (4)$$

In Fig. 2, the spikes in the plot of raw brainwave represent blinks. As the system selects a command based on blink, it does not require any separate classifier. This is a remarkable difference of the proposed system with existing ones.

D. Procedure

At every process cycle (default process frequency is 1 Hz) the device does following things

- 1) System is activated based on the condition stated in Sub-section II-B and the system enters into input sequence.
- 2) The input sequence is consisted of a selector grid of programmable dimension and a moving cursor as shown in Fig. 5. Each block in the grid is mapped to a number like 1, 2, 3 etc. The cursor moves from one block to another at some specific time interval, Δt and selects one block at a time using time-division-multiplexing technique.

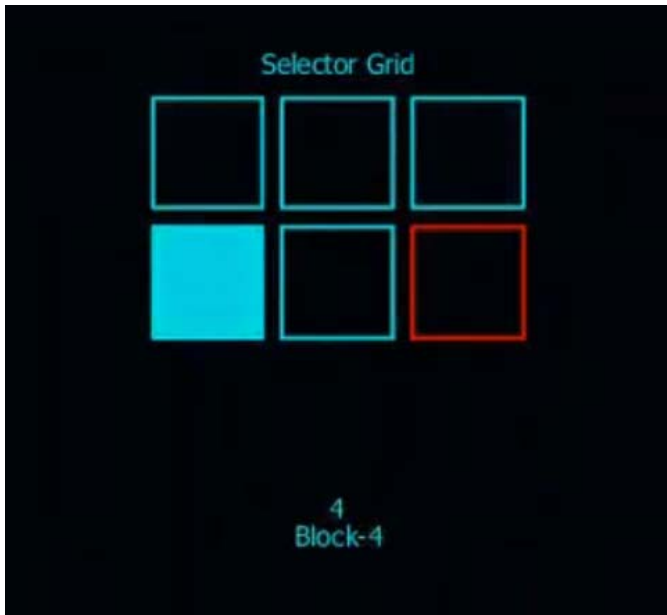


Fig. 3. Selector Grid

- 3) When the user makes a strong blink then the cursor gets locked onto the block it has selected at that moment and waits for a confirmation blink. If the user doesn't provide a confirmation blink then the cursor again starts moving. How the detection of blink works has been described in Sub-section II-C.
- 4) When the user provides the confirmation blink, the program sends the number mapped to the selected block to the processor and again starts moving. The processors then act accordingly. One of the blocks is marked with red which is used to switch back to the idle mode or at the beginning of the program.

III. EXPERIMENTAL RESULTS

A. Parameter Settings

In the experiment, several parameters have been used. They have been tuned for optimization. Optimized values of the parameters are set according to TABLE 2.

B. Accuracy

With considerable help from National institute of traumatology and orthopedic rehabilitation, Bangladesh, the system is experimented on 100 people of different age and sex who had physical disability. The subjects were chosen in a random process. They were given to turn on a specific light using the proposed system from an array of seven lights. The performance of the proposed system is measured in terms of accuracy which is defined as

$$\text{Accuracy} = \frac{\text{Correctly Selected Option}}{\text{Number of Attempts Required}} \quad (5)$$

The result found is shown in bar chart in Fig. 4.

C. Sensitivity Analysis

The performance of the whole process has been found to be sensitive on the proper selection of the system parameters listed in TABLE 2. Accuracy dropped drastically when the value of Δt is decreased further than the optimized value. When the value of the same parameter is increased, the system became too slow although the accuracy increases. Tuning of the weight parameters w_1 and w_2 affects the threshold for attention or meditation and blink which eventually affects the overall accuracy. For larger value of N and hence for larger window size, accuracy slightly deteriorates.

IV. SCOPE AND HUMANITARIAN IMPACT

Anyone can use the device for controlling any digital system, but is best suited for people who are completely physically disabled and cannot even communicate properly by using voice or any other organ but can think or imagine in a particular pattern (cognitive command).

The application of 'Brain-Drive' includes controlling wheelchair, home appliances and even industrial machines using only brainwave (cognitive command), especially for those who suffer from partial or full body paralysis with their brain being the only functioning organ. This will add the impaired people to the main workforce of a country. They can be independent while moving from one place to another. The device also opens a new way of controlling machineries in a new and comfortable way. The expandable output of 'Brain-Drive' gives it reliability, sustainability and acceptability.

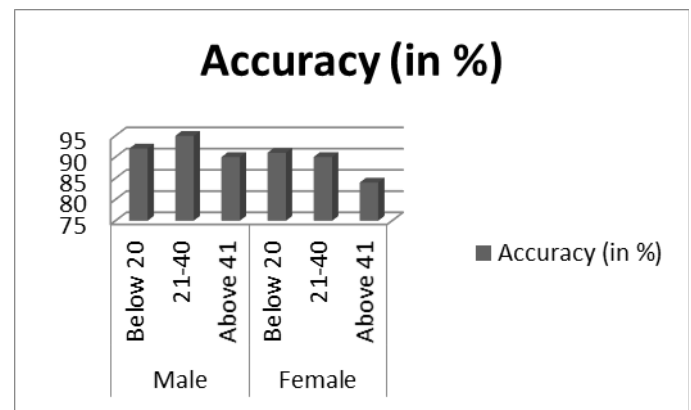


Fig. 4 Experimental Accuracy Rate

TABLE 2 OPTIMIZED VALUES USED FOR DIFFERENT PARAMETERS USED IN THE PROCESS

Parameter	Optimized Value
Δt	2s
w_1	65
w_2	70
N	54

V. CONCLUSION

Brain Computer Interface is one of the emerging technologies that can give the paralyzed people motion. Using the signal brain emits to blink eye, any numbers can be generated using this program. It is more accurate and reliable than other processes as it does not require any classifier. The generated number can be used to control any digital appliances. Most important is that, a very high accuracy is found in our extensive experimentation on several people, both male and female of different age groups, which is very rare in case of EEG based BCI systems. Apart from accuracy, the proposed system offers ease of interfacing and multi-tasking capability. As a whole, the target of developing an EEG based handy device like pen drive for enhancing the use of BCI is successfully fulfilled. The proposed system has the potential to commercialize and spread it to every corner of this world for the benefit of physically challenged people.

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